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Deciphering the Makhul Formation: Overcoming Geological and Geomechanical Challenges to Unlock Reservoir Potential

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Abstract

The unconventional Makhul Formation in Minagish field west Kuwait was evaluated and stimulated for the first time, requiring an integrated geological, geomechanical and stimulation approach to optimize well placement and acid fracturing performance. The formation is characterized by a high-stress strike-slip to reverse fault regime, carbonate heterogeneity, and low permeability (0.1–1 mD). This study aims to assess acid fracturing effectiveness, near-wellbore behavior, and fracture-reservoir connectivity using geomechanical modeling, advanced fluid systems, and data-driven stimulation analysis for future field development in west Kuwait.

A comprehensive geomechanical evaluation was conducted, incorporating rock mechanics testing, stress analysis, and 1D mechanical earth modeling (MEM). The stimulation workflow involved 12 open-hole multistage completion (OHMSC), deploying Single-Phase Retarded Acid System (SPRAS) as the primary extended-contact acid system for deep wormhole penetration and fracture conductivity enhancement. Pressure decline diagnostics in Data Frac analysis provided insights into fracture behavior, fluid leak-off, and reservoir pressure response. The treatments were executed at 30 barrel per minute (bpm), maintaining surface treating pressures below 12,000 psi, ensuring efficient fracture propagation and maximizing reservoir contact across the Makhul Formation.

Ten out of twelve planned acid fracturing stages were successfully completed, covering most of the Makhul Formation and demonstrating good hydraulic fracture connectivity with the reservoir. Post-stimulation coiled tubing milling confirmed full-bore access, followed by nitrogen lifting, which yielded unprecedented production results from the formation. Geomechanical analysis confirmed high stress anisotropy, with an estimated closure pressure of 11,034 psi and a reservoir pressure of 7,400 psi. The low fracture fluid efficiency of 36% indicated significant fluid loss to natural fractures, necessitating advanced fluid control strategies in future treatments.

The fracture compliance effect was observed during pressure decline analysis, suggesting partial fracture closure and width changes post-treatment. The SPRAS exhibited high dissolution capacity and effective viscous fingering, creating deep conductive fractures, making it suitable for future acid fracturing campaigns

in the field. This study establishes a proven workflow for optimizing acid fracturing in highly stressed carbonate formations, enabling the development of the Makhul reservoir as a new production target in west Kuwait's Minagish Field.

This case study represents the first-ever stimulation of the Makhul Formation, integrating geomechanical modeling, acid fracturing, and data-driven analysis to overcome reservoir challenges. The findings provide key insights into fluid behavior, stress anisotropy, and fracture propagation, offering a validated methodology for unlocking unconventional carbonate reservoirs. The success of this treatment sets a benchmark for future acid fracturing strategies and field development in Kuwait's deep, highly stressed formations.

Introduction

The Makhul Formation, located within the Minagish Field in West Kuwait, represents a previously untapped unconventional carbonate interval. It lies stratigraphically between the Tithonian unconformity and the lower Minagish Member, deposited during the Late Jurassic (Tithonian) to Early Cretaceous (Berriasian) periods in a low-energy inner to outer ramp marine setting. This formation exhibits significant heterogeneity and complexity, with deposition influenced by relative sea-level fluctuations and third-order transgressive-regressive cycles superimposed on a second-order eustatic trend.

Recent advances in geomechanical evaluation, stimulation chemistry, and completion technology have created new opportunities to assess and unlock the potential of tight carbonate reservoirs such as Makhul. This paper presents the first-ever stimulation campaign in the Makhul Formation, demonstrating how integration of geological modeling, mechanical earth modeling (MEM), and advanced acid fracturing workflows enabled successful well placement and reservoir access.

The primary objective of this study is to evaluate stimulation effectiveness and reservoir behavior in this low-permeability, high-stress carbonate interval. Specific aims include:

- Characterizing the geomechanical environment and stress regime through rock mechanics and MEM analysis,
- Designing and executing a multistage acid fracturing treatment using Single Phase Retarded Acid System (SPRAS),
- Interpreting pressure diagnostics to understand near-wellbore fracture development and reservoir connectivity,
- Evaluating production outcomes to benchmark future Makhul development strategies.

This effort represents a milestone in Kuwait's unconventional reservoir stimulation, establishing a workflow for developing similar tight, highly anisotropic carbonate plays.

Geological and Petrophysical Overview of the Makhul Formation

The Makhul Formation has long been recognized as a source-prone interval but only recently re-evaluated as a viable reservoir target due to advances in unconventional carbonate stimulation.

Depositional Setting and Stratigraphy

Stratigraphically, Makhul sits above the Jurassic Tithonian unconformity and beneath the Minagish Lower Member, deposited during transgressive-regressive cycles as shown in Figure 1.

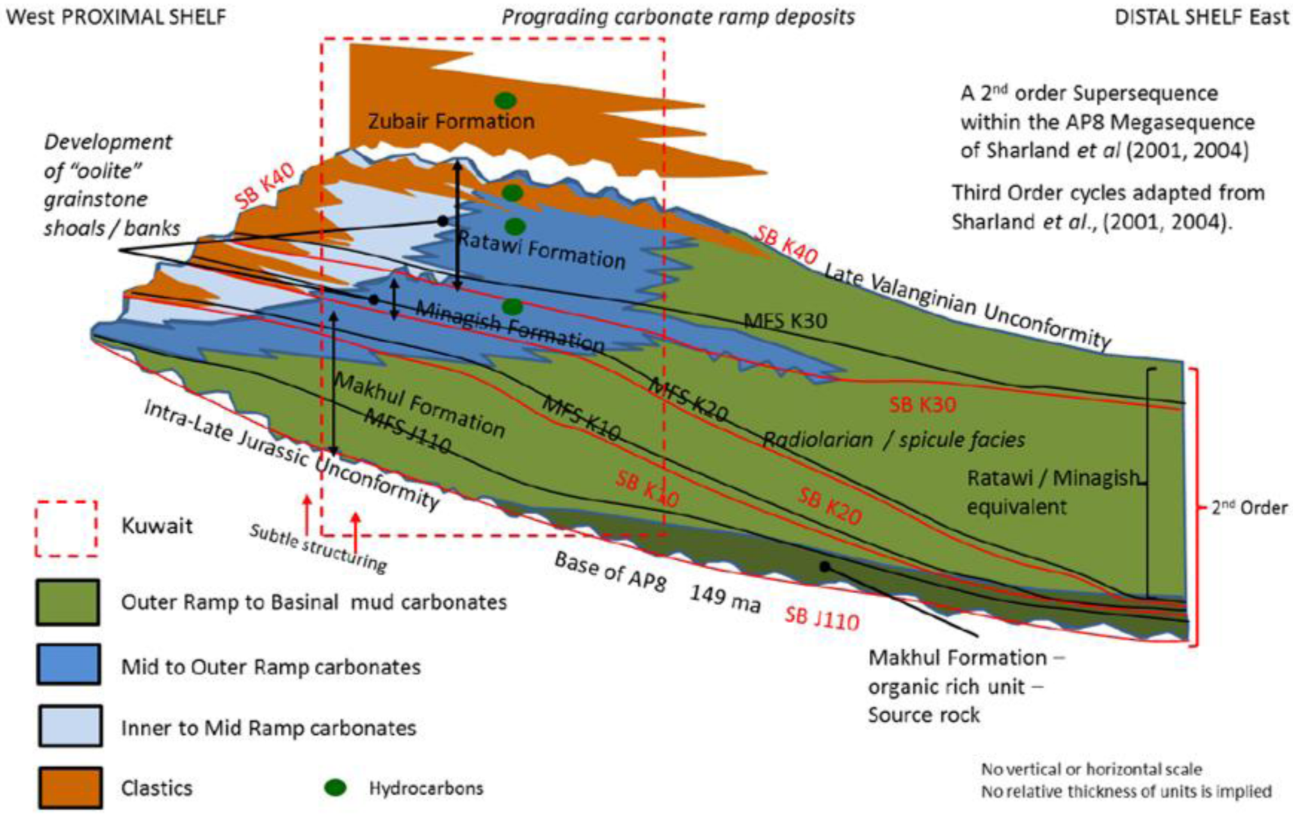

Figure 1—Stratigraphic Column Highlighting Makhul Position and Key Markers

Makhul was deposited in a low-energy carbonate ramp environment, transitioning from proximal to distal marine facies. Regional chronostratigraphy from the late Tithonian to mid-Valanginian shows significant facies variability and compartmentalization, driven by relative sea-level variations and fault block tectonics. These depositional factors have contributed to the heterogeneity observed in both matrix and fracture properties. Depositional modeling indicates lateral and vertical heterogeneity driven by relative sea-level fluctuations and fault block isolation.

Petrophysical Properties and Reservoir Heterogeneity

Key petrophysical attributes of the Makhul Formation include:

- Low matrix porosity and permeability: Average matrix porosity is typically in the range of ~0.015%, with matrix permeability well below 1 mD. Productive capacity relies heavily on fracture development.
- Fracture-driven storage and flow: Effective fracture porosity from core analysis is consistent with values obtained from dual-porosity modeling, confirming the dominance of secondary porosity mechanisms.
- Mechanical layering and anisotropy: Horizontal permeability significantly exceeds vertical due to mechanical stratification. Effective lateral permeability can locally exceed 200 mD, especially in zones of structural curvature.
- Fracture orientation sensitivity: Maximum horizontal permeability is aligned NE-SW (N20°E–N40°E), favoring fractures and faults oriented in these directions while hindering flow along NW-SE or E-W alignments.
- Fault-dominated drainage geometry: The Makhul structure in the Minagish Field is segmented by major faults, resulting in large drainage blocks with uncertain interconnectivity and pressure support.

Core and log calibration confirmed reliance on fracture-driven storage and flow.

Fracture porosity modeling shows higher NE–SW directional conductivity aligned with interpreted maximum horizontal stress as observed in Figure 2. AI-driven static models were used to extrapolate porosity and fracture intensity across data-scarce regions.

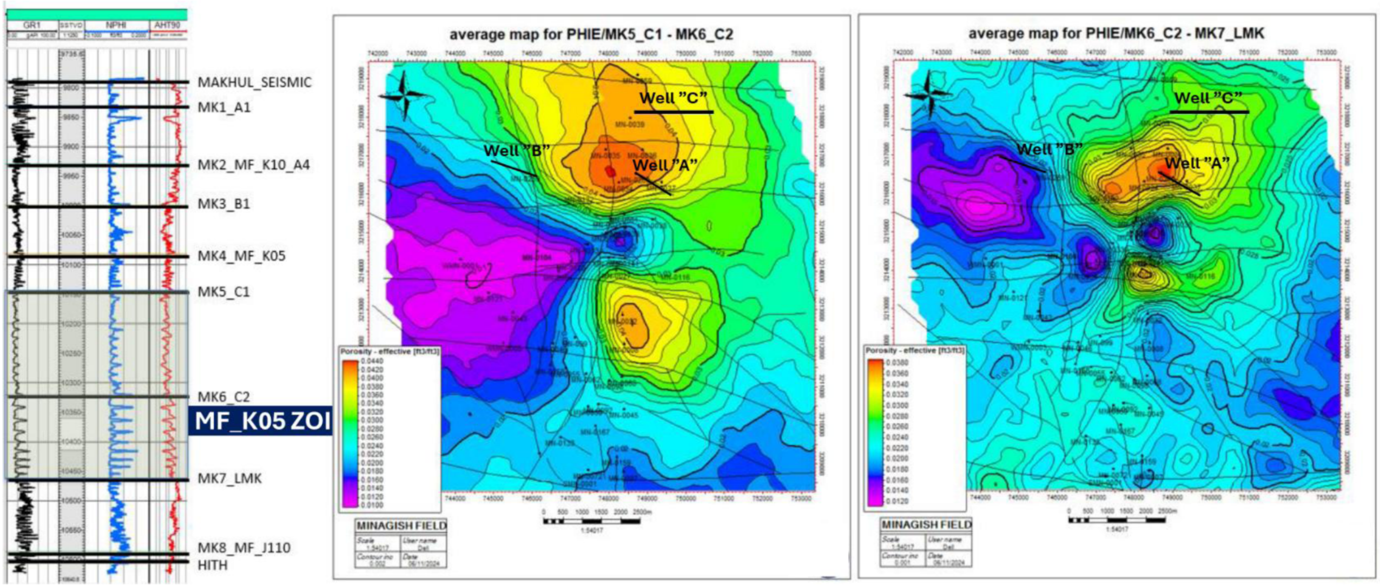

Figure 2—Fracture Porosity Map with Curvature Anomalies and Permeability Direction

Key Development Uncertainties

Development of the Makhul Formation is challenged by several geological uncertainties:

- Distribution and continuity of matrix properties outside well control,
- Fracture size, intensity, and orientation beyond borehole observations,
- Sealing behavior of major faults and lineaments,
- Drainage volume estimates, which assume ~50 ft effective radius per fracture in the Distributed Fracture Network (DFN) model.

These uncertainties influence forecasts related to long-term production sustainability, aquifer support, pressure maintenance, and oil-water contact (OWC) depth positioning. Vertical appraisal wells with MDT surveys are recommended to reduce these uncertainties and calibrate horizontal well placement and stimulation design.

Figure 3 shows the structural fault framework overlaid with 2 exploration Makhul wells, A and B, based on which data have been gathered to fully develop Well C, the subject well of this paper.

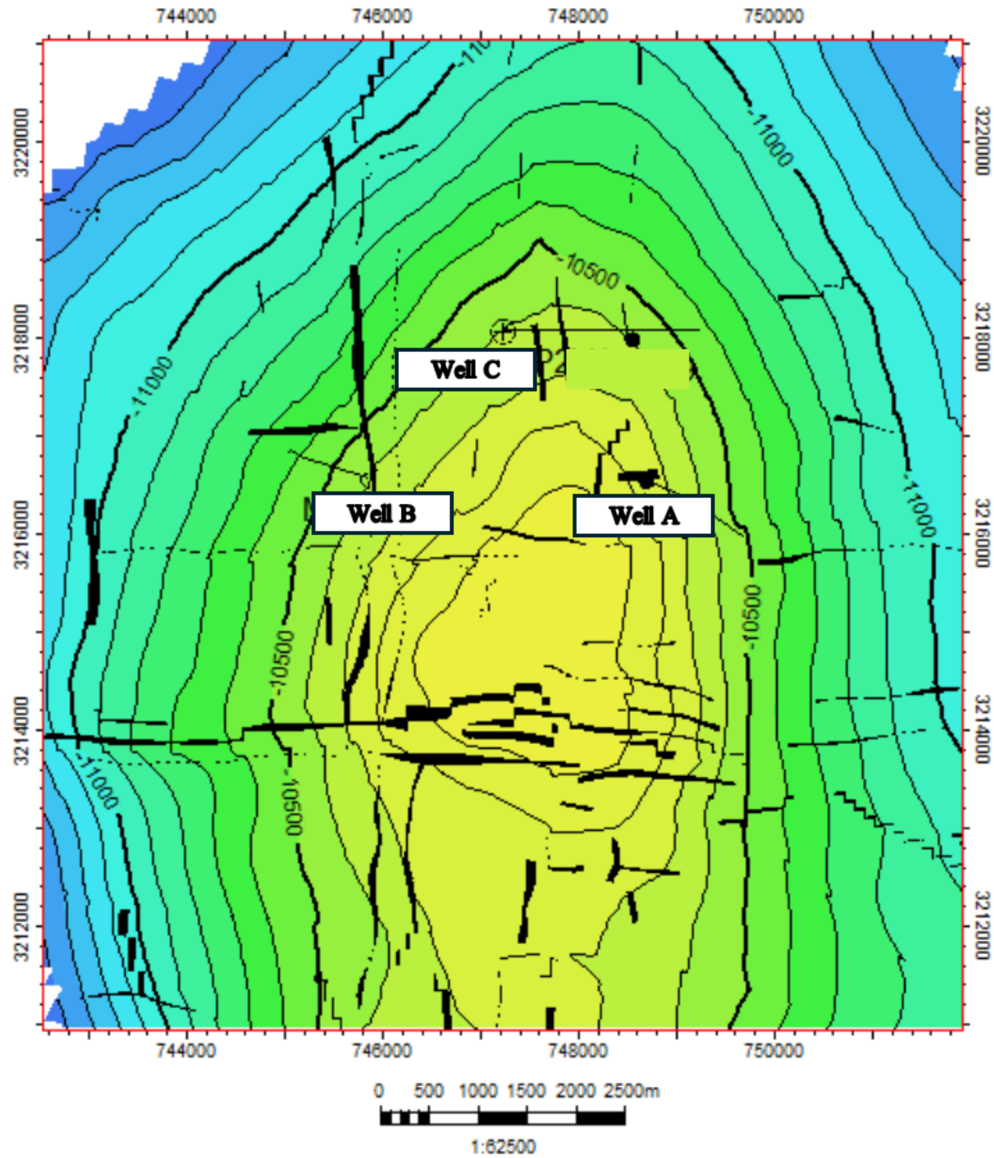

Figure 3—Structural Fault Framework Overlaid with Prospective Deviated/Horizontal Well Paths

Reservoir Properties Measurements and Analysis

Figure 4 shows the open hole logs (resistivity, neutron, density and GR) along with caliper and pressure test results. Average neutron porosity of around 2 – 4 % was observed across Makhul in Well-B.

Downhole formation testing was utilized to acquire reservoir pressure measurement. Since very low mobility was expected, large inlet area – dual packer – was utilized accordingly. All tests showed very tight or dry test behaviors.

The below figure 5 is the only test where decent reservoir pressure build up was observed. Reservoir pressure was building up to around 7200 psia after around 2 hrs of buildup revealing a mobility that is less than 0.01 md/cp. Pressure was still building up gradually when the test was aborted. Possible reservoir pressure supercharging is expected as well in such low mobility formations.

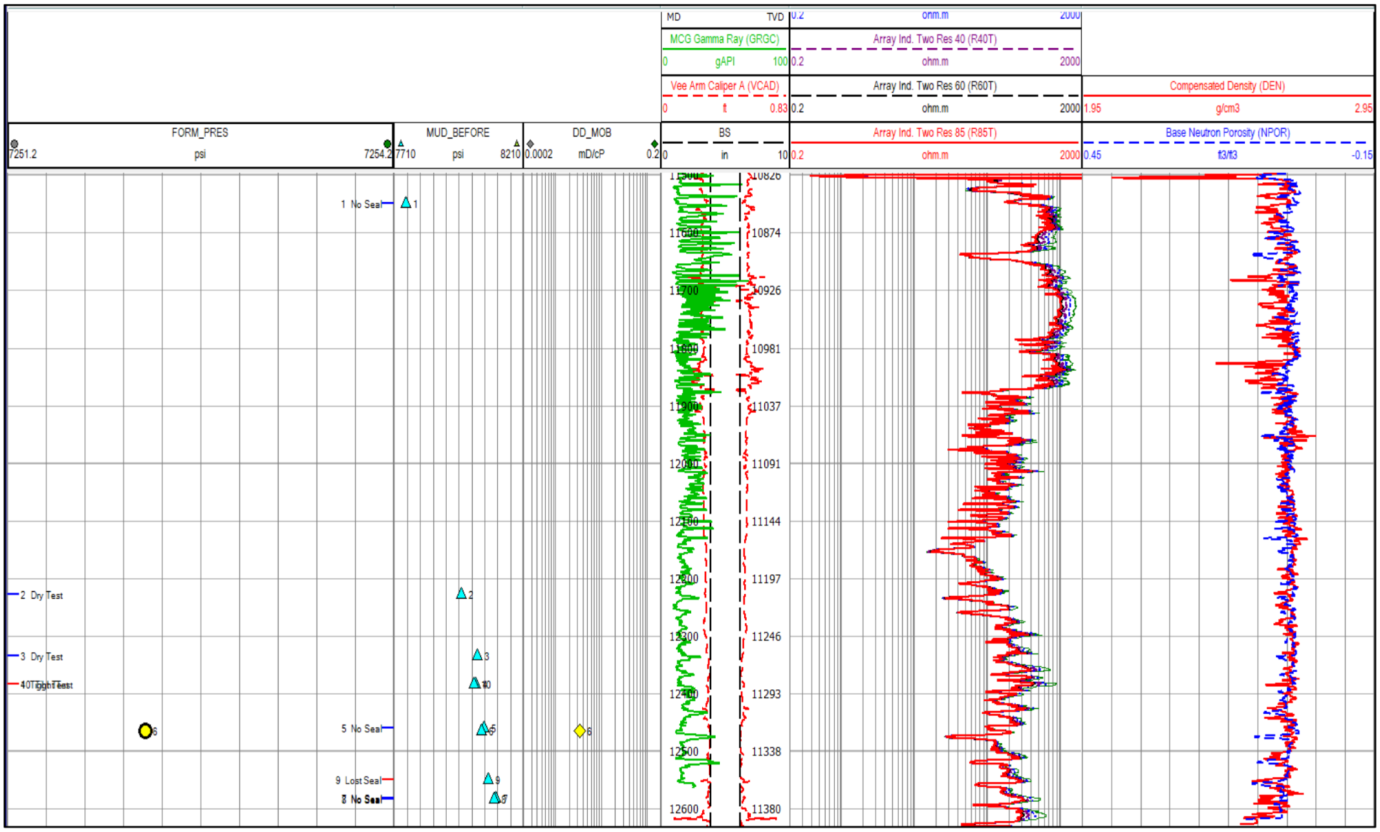

Figure 4—Open hole logs with pressure profile

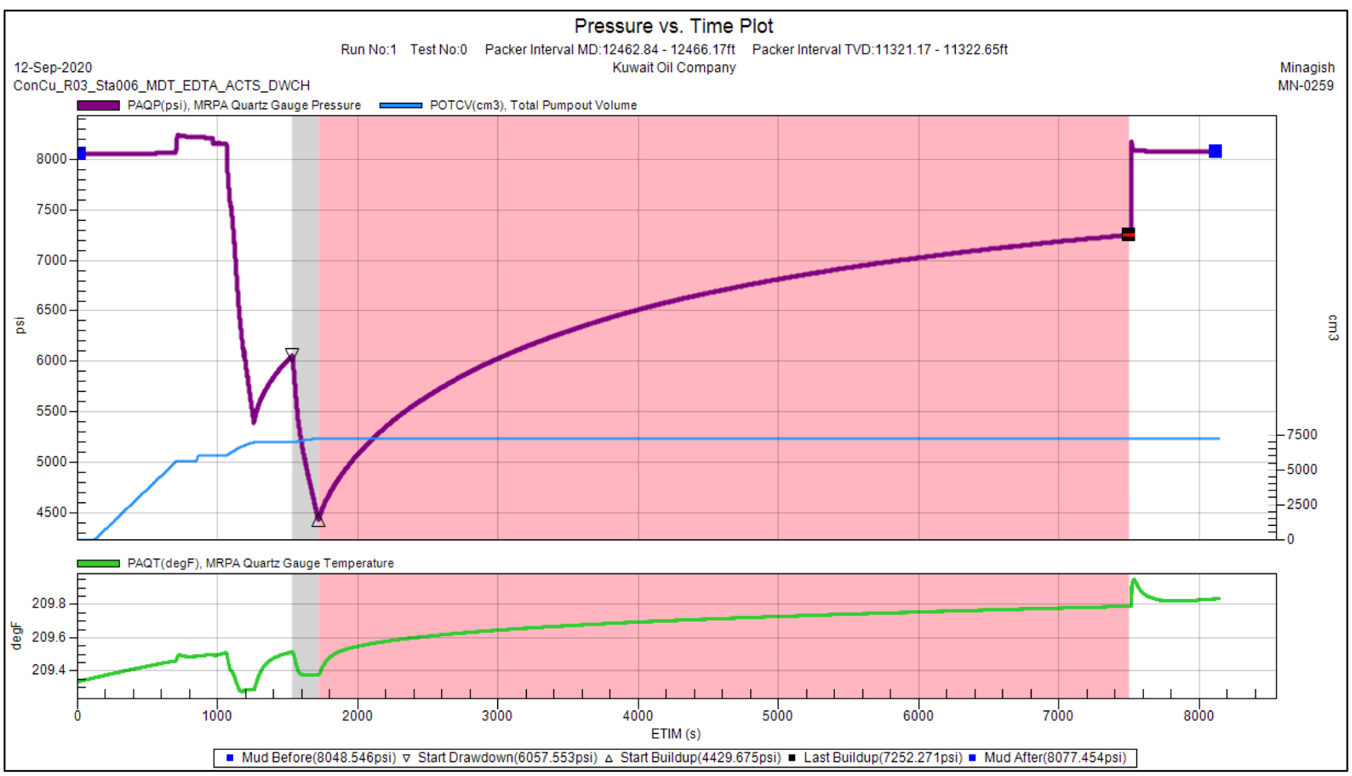

Figure 5—Pressure test at 12463 – 12466 ft MD

Below figure 6 shows dry test behavior at a different interval. It can be observed that tiny amounts of volume withdrawal cause significant pressure drop with almost no pressure build up. Most of the points across Makhul in Well-B showed the same behavior.

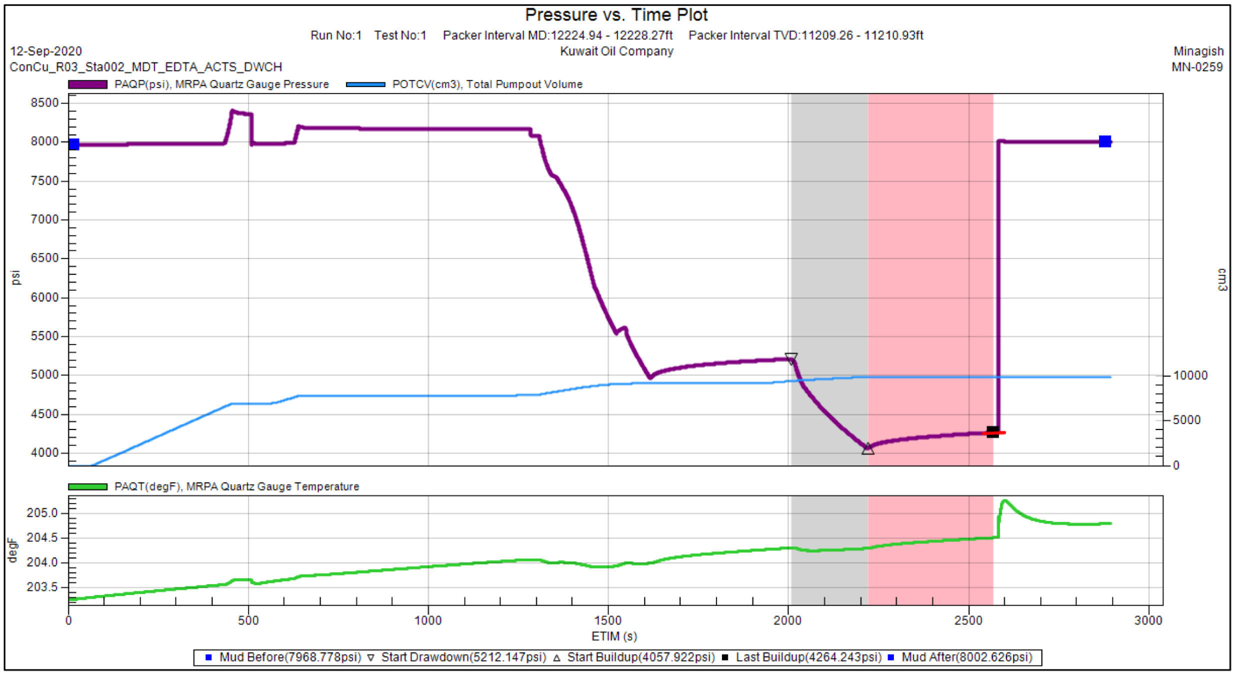

Figure 6—Pressure test at 12225 – 12228 ft MD

Figure 7 below shows tight / very tight reservoir mobility response where very slow pressure build up is observed.

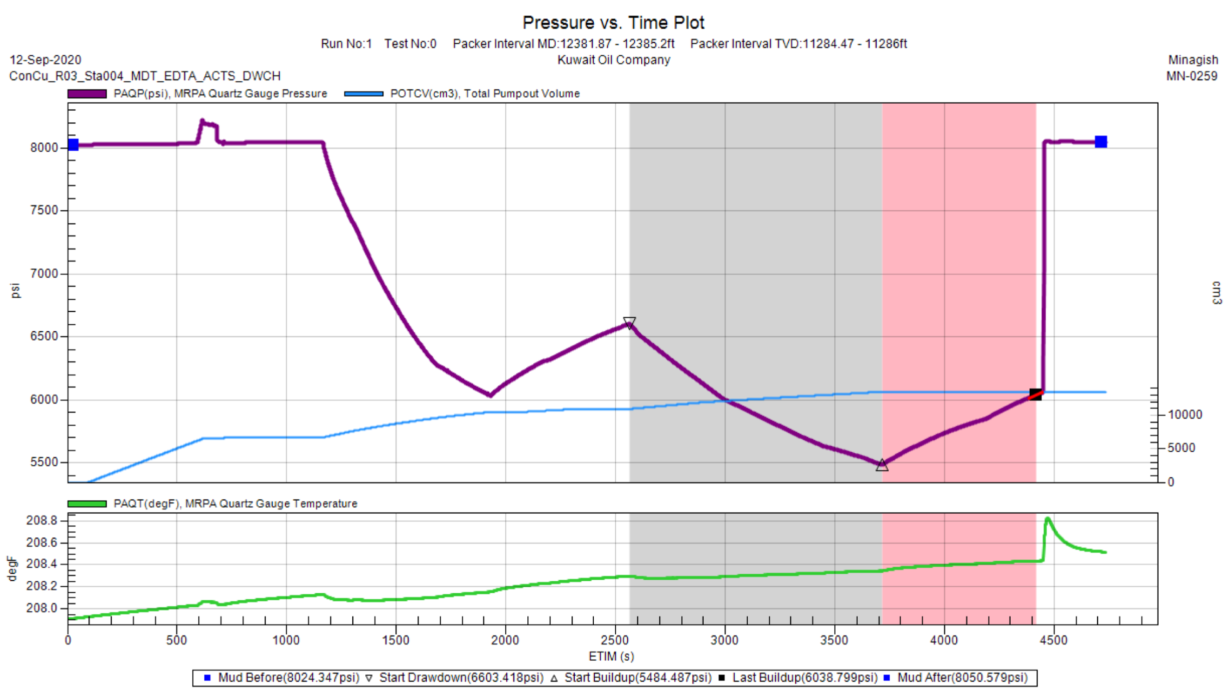

Figure 7—Pressure test at 12382 – 12385 ft MD

The above raw pressure test data demonstrates that Makhul formation is ultra tight / unconventional from mobility / permeability point of view.

Geomechanical Analysis with Rock Testing

Makhul formation presents unique challenges related to fracture initiation, propagation, and long-term conductivity. Acid fracturing aims to enhance well productivity by creating fractures that increase the flow of hydrocarbons, and geomechanical modeling provides the necessary insights into the stress state, rock mechanics, and fracture behavior to optimize this process. Geomechanical modeling was carried out with the following objectives:

- Identify stress barriers that limit vertical and lateral fracture growth
- Provide input information for designing treatments that minimize fracture closure and maximize long-term conductivity

Geomechanical modeling incorporates key rock mechanical properties such as: Young's modulus (rock stiffness), Poisson's ratio (rock deformability), tensile strength and cohesion. Laboratory measurements of rock properties are crucial for validating models and designing acid fracturing treatments that can overcome the formation's mechanical resistance. MF-05 Zone of Interest was well-represented by the core from the well B, while the overlying Makhul layers were not cored.

Rock mechanics testing comprised triaxial compression tests to evaluate in-situ mechanical properties, Brazilian tensile strength tests, and Biot coefficient tests for poroelastic characterization. Prior to mechanical testing, the estimated in-situ stress state for each sample was calculated as the mean of the vertical and horizontal stresses minus the pore pressure. It is well established that elastic moduli are pressure-dependent, with more porous and compliant rocks exhibiting greater sensitivity to confining pressure changes (Somogyi Molnár *et al.*, 2014). To ensure mechanical properties reflect near in-situ conditions, measurements were interpreted with respect to these estimated effective stresses. The in-situ static Young's modulus was sometimes comparable, sometimes 25% higher than the modulus measured at the close to zero confining pressure stage. This trend underscores the importance of stress conditioning in laboratory tests to replicate in-situ mechanical behavior accurately (Figure 8).

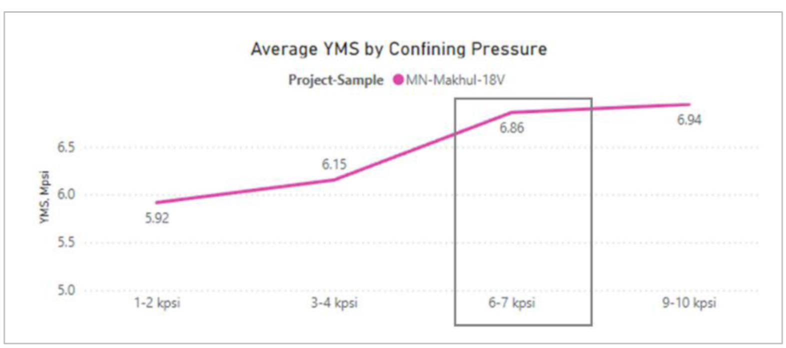

Figure 8—An example of static Young's modulus change with increase of confining pressure

The in-situ static Young's modulus is approximately 6-8 Mpsi, which is 20-30% lower than dynamic values. Poisson's ratio static is very close to the dynamic values. Biot's constant varies from 0.41 to 0.81. All these mechanical properties are shown in Figure 9. Low values indicate that the mechanical response is primarily controlled by its solid matrix rather than the pore fluid.

Core Interval (ft)	Density Bulk, TXC	Pc, psi	YMS in situ, Mpsi	PRS in situ	FA, deg	UCS from TX, psi	TSTR, psi
12017	2.68	6100	7.58	0.32	37	32456	
12026	2.65	5400	6.02	0.34	39	12936	1197
12056	2.67	6200	6.86	0.31	38	4153	810
12129	2.66	5900	8.04	0.28	31	30867	1352
12145	2.69	5500	8.13	0.28	36	34309	1833
12157	2.67	5200	7.86	0.29	33	38277	1930
12225	2.60	4684	6.88	0.27	31	33432	1230
12324	2.68	5900	8.70	0.28	42	25861	
12383	2.74	6338	9.30	0.31	37	34240	
12393	2.67	6338	7.14	0.29	44	20354	
12407	2.66	6338	6.68	0.31	30	16782	

Figure 9—Mechanical parameters measured on core indicate that Makhul rock is dense, stiff, and strong

Unconfined Compressive Strength (UCS) measurements for the Makhul Formation yielded an average value of approximately 31,000 psi, indicating that the formation comprises very strong rock. However, a subset of samples exhibited significantly lower UCS values, with some as low as 4,000 psi—well below the formation average. To investigate these anomalies, high-resolution photographs and computed tomography (CT) scans were employed as shown in Figure 10 and Figure 11. These analyses revealed that intact samples consistently exhibited UCS values in the range of 20,000 to 30,000 psi, while samples containing internal microfissures or structural defects displayed markedly reduced strength. As a result, samples identified as damaged or fractured based on visual inspection and CT imaging were excluded from the final UCS dataset to ensure that the reported mechanical properties accurately represent the intact rock fabric of the Makhul Formation. Tensile Strength averaged at 1470 psi, which is around 5% from Unconfined Compressive Strength.

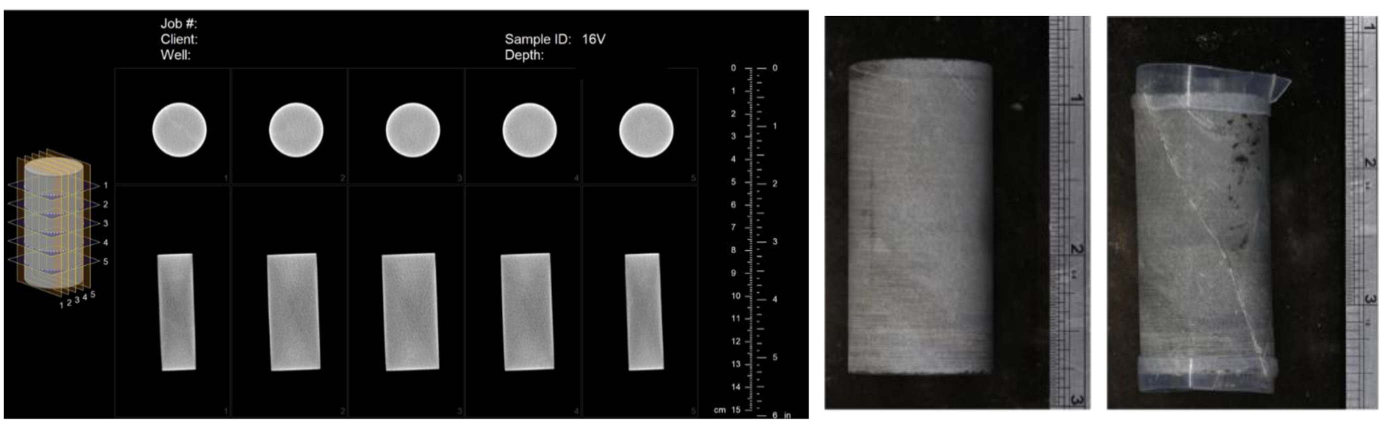

Figure 10—An example of an intact sample without pre-existing microfractures (#32,000 psi UCS). Left: CT scan prior to the test. Right: Photos taken before and after the test.

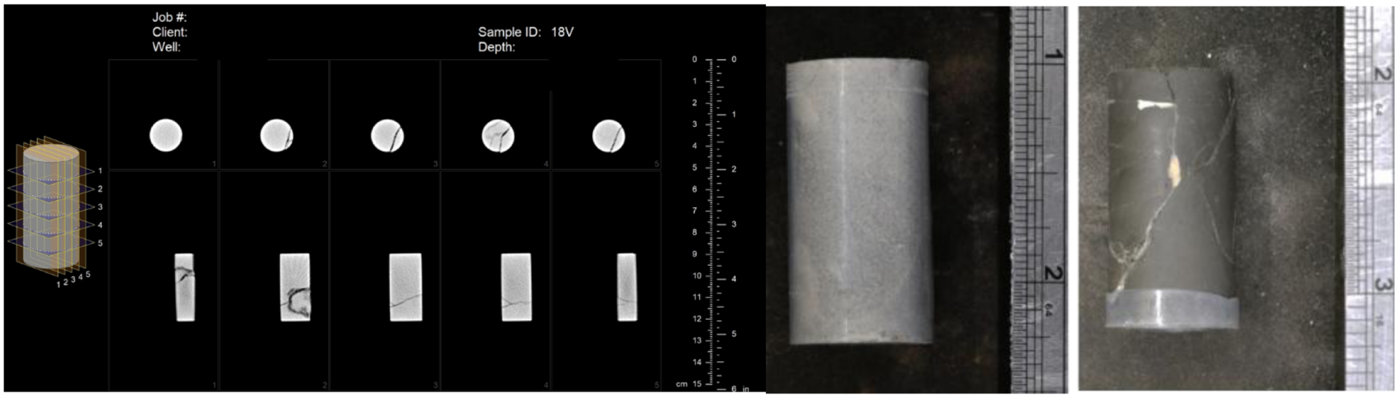

Figure 11—An example of a sample with pre-existing microfractures (#4,000 psi UCS). Left: CT scan prior to the test. Right: Photos taken before and after the test.

In addition to rock mechanical properties, accurate characterization of the in-situ pore pressure and stress state is essential for any robust geomechanical model. These parameters govern the effective stress conditions, influence rock failure criteria, and directly impact the prediction of fracture initiation, propagation, and overall reservoir behavior under production or stimulation.

Pore pressure in the Makhul Formation is interpreted to be supra-hydrostatic based on drilling event analysis and limited available XPT data. The degree of overpressure varies across the field, with equivalent mud weight estimates ranging from 11 to 15 ppg, indicating significant spatial variability in pore pressure regimes. Direct stress measurements are limited; however, a single breakdown test and closure pressure analysis were conducted in Well B. The closure pressure gradient in this well was estimated at approximately 0.97 psi/ft, as shown in Figure 12. This value closely approaches the calculated vertical stress gradient, suggesting that the formation is in a highly stressed state, possibly near lithostatic conditions.

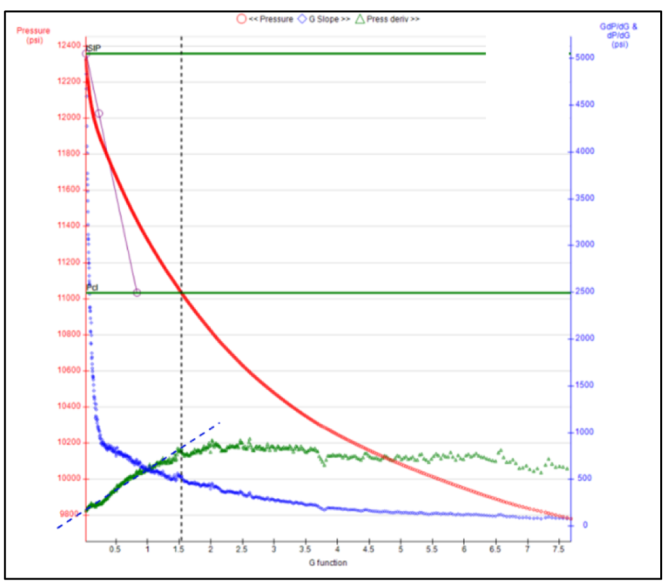

Figure 12—Closure pressure gradient in well B was estimated 0.97 psi/ft. The value approaches the gradient of vertical stress, suggesting highly stressed conditions.

Drilling event analysis was performed across 13 wells with varying trajectories and mud weight (MW) strategies. Indications of formation pressure—such as gas shows, kicks, and inflows—were observed in multiple wells within the Makhul Formation at MWs ranging between 12.5 and 15.0 ppg. Minor losses were reported in three of the wells when MW ranged from 13.4 to 14.7 ppg, potentially indicating that the mud pressure exceeded the local fracture pressure or that the well intersected a conductive fault network. Notably, both Wells B and C penetrated heavily faulted intervals but did not exhibit any losses, suggesting localized heterogeneity in fault permeability or sealing behavior.

Wellbore stability issues were also recorded: tight hole conditions and borehole cavings occurred in three wells with MWs between 13.2 and 14.0 ppg. However, no consistent correlation was observed between wellbore deviation and instability, implying that trajectory alone was not a dominant factor.

Borehole image logs shown in Figure 13 from several wells revealed the presence of well-developed shear failures (borehole breakouts) within the Makhul Formation. These features are consistent with hole enlargement and overpull events driven by inadequate mud weight to counterbalance in-situ formation stresses. Breakouts were particularly pronounced in the upper and lower intervals of the Makhul, indicating lower mechanical stability relative to the middle section. The observed hole ovalization consistently occurred in a NW-SE direction (~135°), aligning with the minimum horizontal stress orientation. This directional enlargement confirms stress-driven instability and points to significant stress anisotropy.

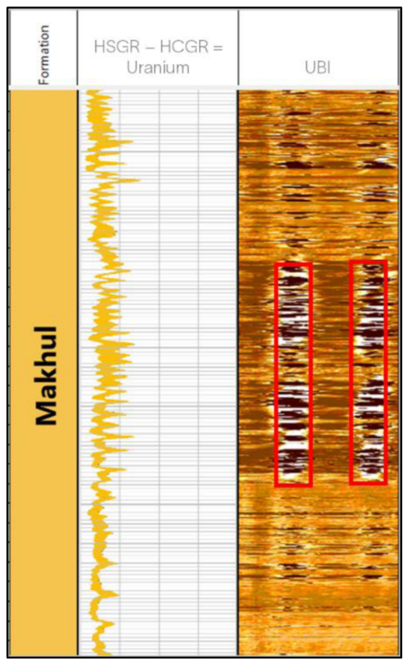

Figure 13—Image data reveals shear failures (breakouts) in Makhul

Importantly, the presence of high unconfined compressive strength (UCS) values—frequently exceeding 25,000 psi—combined with evidence of shear failure under drilling conditions, strongly suggests that substantial differential stress is required to induce failure in such strong rock. This underscores the role of pronounced in-situ stress anisotropy in governing wellbore stability within the Makhul Formation.

Another interesting observation indirectly suggesting the stress state are small features on core (Figure 14) —namely, horizontal to vertical stylolites. These dissolutions provide valuable clues about the evolution of the stress field. Vertical stylolites typically form perpendicular to the maximum compressive stress (σ), which in this case would have been horizontal, indicating tectonic compression (Figure 15). In contrast, the presence of earlier horizontal stylolites suggests an earlier phase where σ was vertical, consistent with burial-related compaction. The observed transition from horizontal to vertical stylolites therefore records a shift in the stress regime—from vertical compaction during diagenesis to horizontal compression during later tectonic deformation. This progressive change, coupled with the development of shear fractures, supports a model of increasing differential stress culminating in brittle failure.

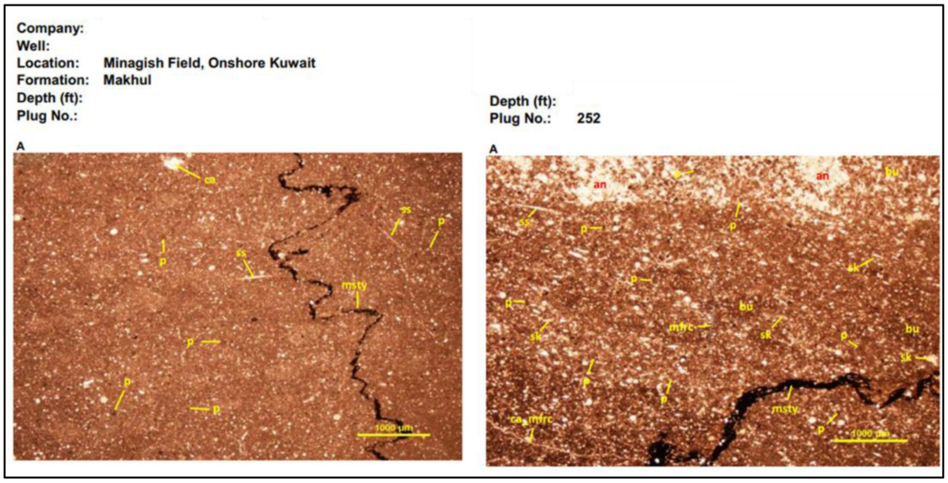

Figure 14—Horizontal to vertical stylolites present on core from Makhul

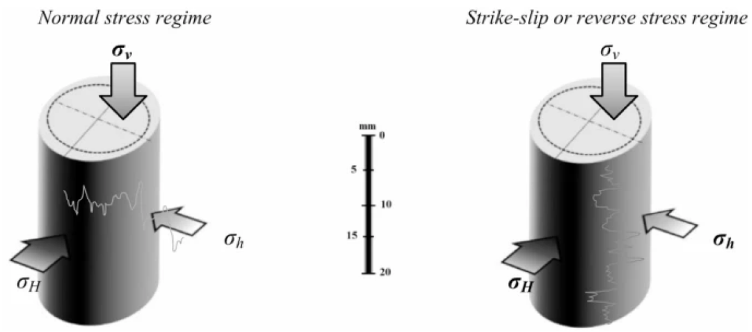

Figure 15—A simple schematic explanation of the genetic classification of stylolites based on the stress regimes. The head seam of stylolites is in the direction of the maximum main stress [Al-Yaseri, 2024]

The required log data for mechanical earth modeling were available for both offset wells and the target Well C. After performing quality control and calibration, it was observed that Well C exhibits a more brittle lithological profile compared to offset Well B, as shown in Figure 16. This increased brittleness suggests a greater tendency for fracture propagation under stimulation, potentially reducing the required breakdown pressure. Brittle rocks are generally more favorable for hydraulic or acid fracturing because they fail more readily by fracture rather than by plastic deformation. They tend to form cleaner, more extensive fractures with better connectivity, which is ideal for reservoir stimulation. However, it does not directly imply lower in-situ stress magnitudes, which are governed by regional geostress conditions and formation depth.

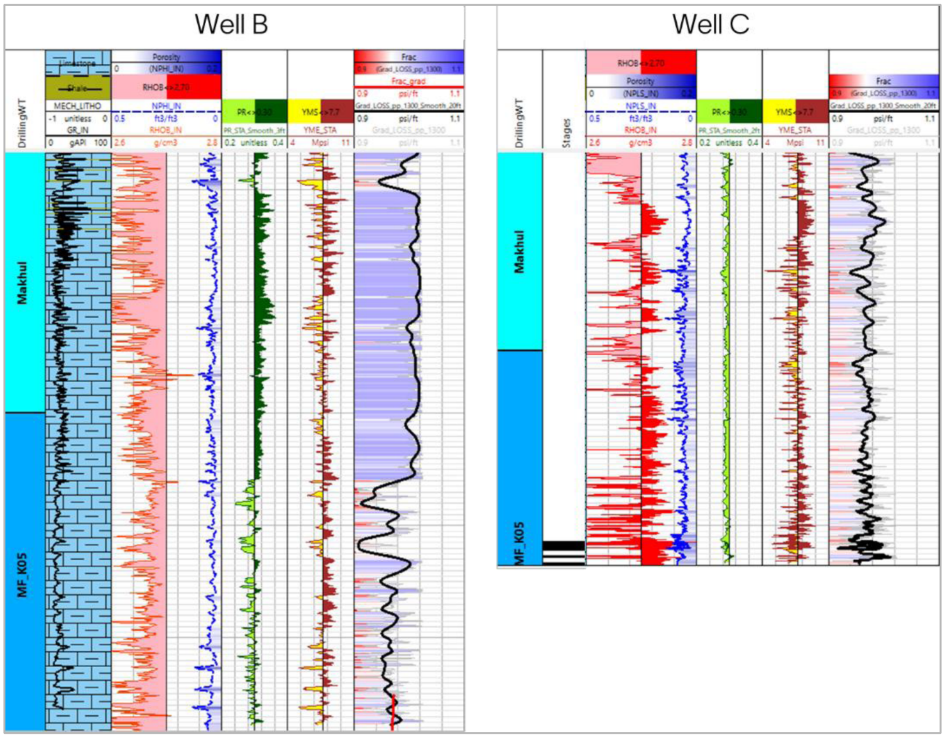

Figure 16—Well C exhibits a more brittle lithological profile compared to offset Well B. The stress gradient is validated against Datafrac results, and the mechanical properties are based on core measurements.

Acid Fracturing Strategy and Treatment Design

With the available data from geology, drilling reports, and open hole logs, and utilizing information from analog formations as well as offset wells and best practices across middle east for Carbonate stimulation the treatment dosage and treatment sequence schedule was based on PAD: Acid 1:1 ratio of average dosage 500 gpf (250:250) across the OH section, each stage consists of 6 cycles of alternating crosslinked PAD for Fracture initiation and geometry propagation followed by lightweight single-phase retarded acid system (LWSPRAS) to enhance deep reservoir acid placement enhancing stimulation and flowback effectiveness and Visoelastic Diverting Acid for optimized diversion. The treatment design is observed in Table 1.

Table 1—Stimulation Treatment Design per stage

Step No.	Step Name	Pump Rate	Fluid Name	Step C.F Vol	Cummulative Fluid Volume	Acid Concentration
		(bpm)		(bbl)	(bbl)	(%)
1	Pre-Flush	25	Mutual Solvent	275	275	0
2	PAD-1	25	Crosslinked Gel	200	475	0
3	Main Acid	25	LWSPRAS	180	655	20
4	Diverter	25	VDA-20	110	765	20
5	PAD-2	25	Crosslinked Gel	200	965	0
6	Main Acid	25	LWSPRAS	180	1145	20
7	Diverter	25	VDA-20	110	1255	20
8	PAD-3	25	Crosslinked Gel	200	1455	0
9	Main Acid	25	LWSPRAS	180	1635	20
10	Diverter	25	VDA-20	110	1745	20

Step No.	Step Name	Pump Rate	Fluid Name	Step C.F Vol	Cummulative Fluid Volume	Acid Concentration
		(bpm)		(bbl)	(bbl)	(%)
11	PAD-4	25	Crosslinked Gel	200	1945	0
12	Main Acid	25	LWSPRAS	180	2125	20
13	Diverter	25	VDA-20	110	2235	20
14	PAD-5	25	Crosslinked Gel	200	2435	0
15	Main Acid	25	LWSPRAS	180	2615	20
16	Diverter	25	VDA-20	110	2725	20
17	PAD-6	25	Crosslinked Gel	200	2925	0
18	Main Acid	25	LWSPRAS	180	3105	20
19	Post-Flush	25	Mutual Solvent	275	3380	0
20	CFA	25	20% HCL	40	3420	20
21	Flush	6	Slickbrine	140	3560	0

Based on the simulations and developed 1 Dimension-Mechanical Earth Model (1D-MEM), using a Multiphysics Acid Fracturing model the fracture geometry showed an average 205 ft etched fracture half-length, 206 ft average etched height, and 0.95 inch etched width at wellbore. Figure 17 demonstrates the simulated acid fracture geometry with changes in fracture conductivity.

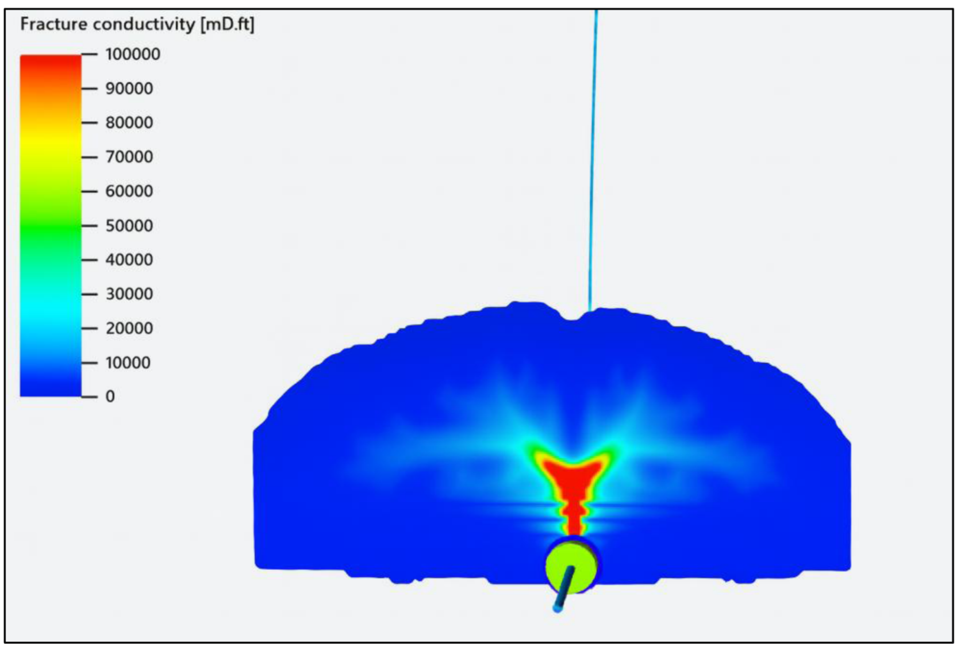

Figure 17—Treatment Fracture Geometry Simulation

Execution Results and Fracture Diagnostics

The acid fracturing campaign for Well C was successfully executed across twelve planned stages using the Open-Hole Multistage Completion (OHMSC) methodology in an openhole horizontal section with mechanical packers used for isolation between stages. Out of the twelve targeted intervals, ten stages were completed successfully, while two were aborted due to operational challenges. The stimulation treatments followed the MSAF (Multistage Acid Fracturing) design, with fluid systems customized per stage.

Pumping Summary and Treatment Pressures

The execution maintained a steady pumping rate of 30 bpm, with surface treating pressures remaining below 12,000 psi across all successful stages. Pressure profiles showed strong fracture initiation and propagation responses. Each stage was monitored in real-time from surface, allowing dynamic response to pressure trends and fluid efficiency observations. Figure 18 shows treatment stages 1 and 3 surface pressure and rates charts.

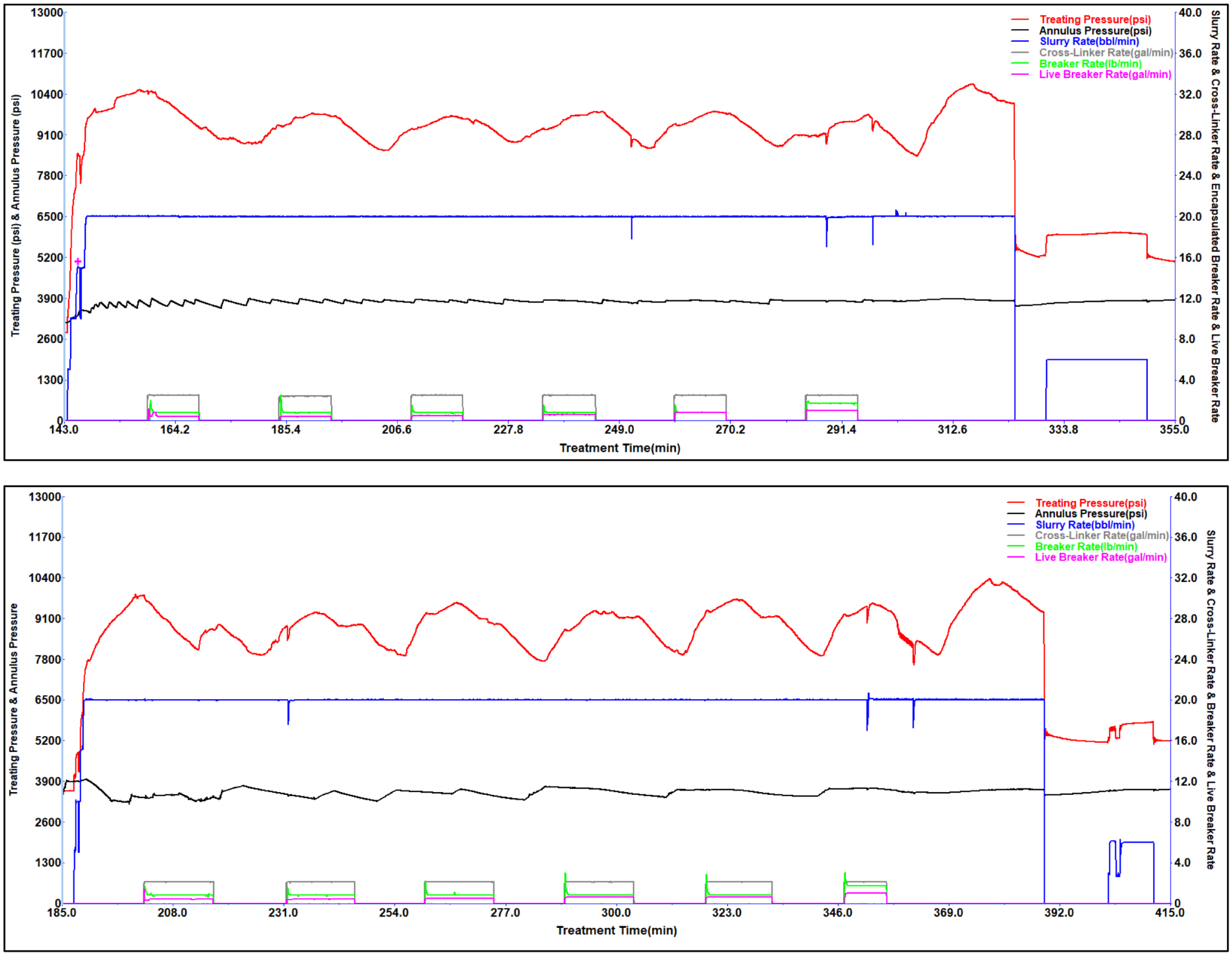

Figure 18—Stage 1 (top) and 3 (bottom) Pumping Surface Pressure and Rate Plots

Pressure Decline Analysis

For each stage, pressure decline curves were analyzed to interpret:

- Fracture extension behavior,
- Fluid efficiency and leakoff characteristics,
- Closure pressure and compliance effects.

A notable fracture compliance effect was observed in multiple stages, evident in the gradual slope change during falloff, suggesting partial fracture closure and width reduction over time. This is consistent with high in-situ stress anisotropy identified in the MEM analysis. Figure 19 shows the Stage 1 breakdown injection test after closure analysis.

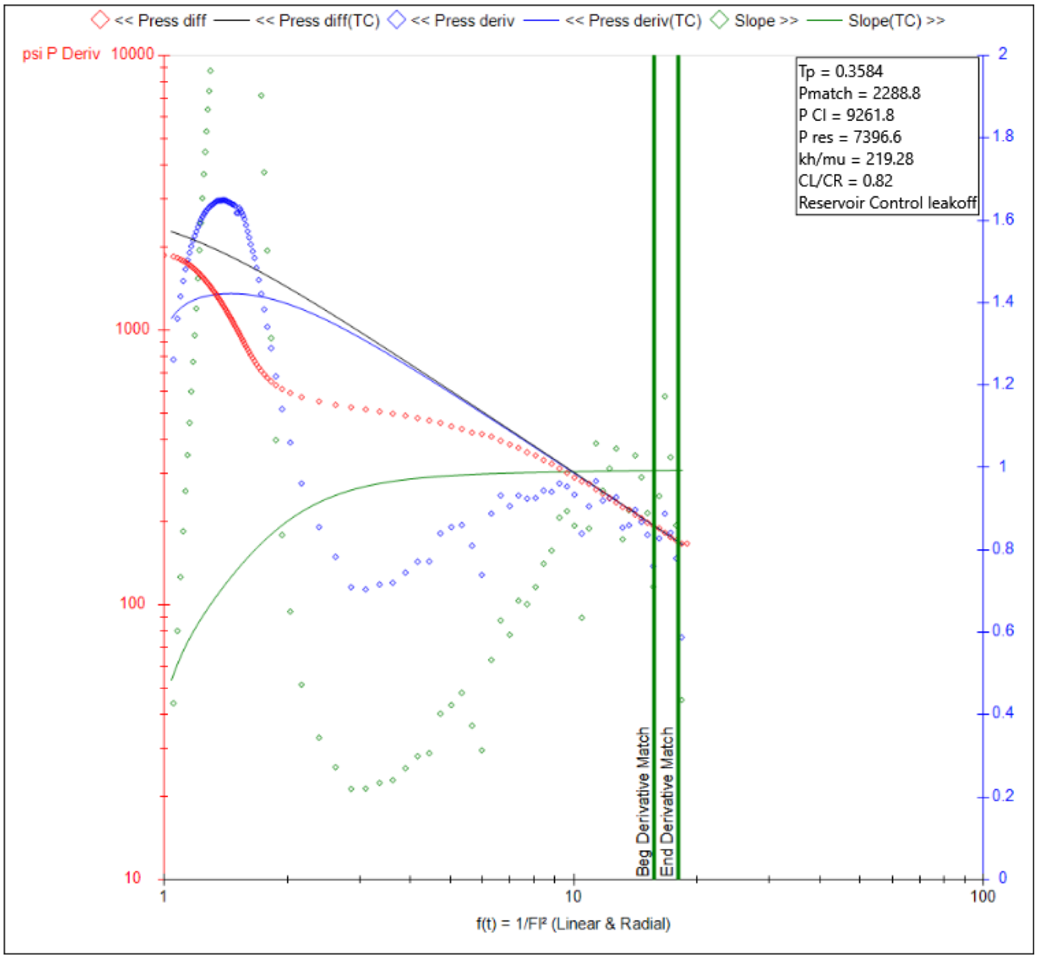

Figure 19—Stage 1 Breakdown Test After Closure Analysis

The average closure pressure was estimated around 11,034 psi, while reservoir pressure post-treatment was determined to be approximately 7,400 psi. Fluid efficiency was measured at ~36%, indicating considerable fluid loss to existing natural fractures which is a common phenomenon in low-permeability carbonate systems with pre-existing discontinuities.

Stage-Wise Observations

Stages 1–6: Performed within expected pressure envelope. Leakoff rates were manageable, and early diagnostics indicated strong pressure recovery.

Stage 7: Encountered unexpected pressure buildup and potential partial fracture containment; operation paused and redesigned for subsequent stage.

Stages 8–12: The pressure and rate profiles remained stable throughout the stages, indicating consistent and controlled treatment execution. Pressure data exhibited no abnormal spikes, confirming effective fracture propagation by the fracturing fluid. Additionally, there was no indication of fluid loss or unexpected formation breakdown beyond the designed parameters.

Figure 20 compares all the treatment plots for the 10 acid fracturing stages.

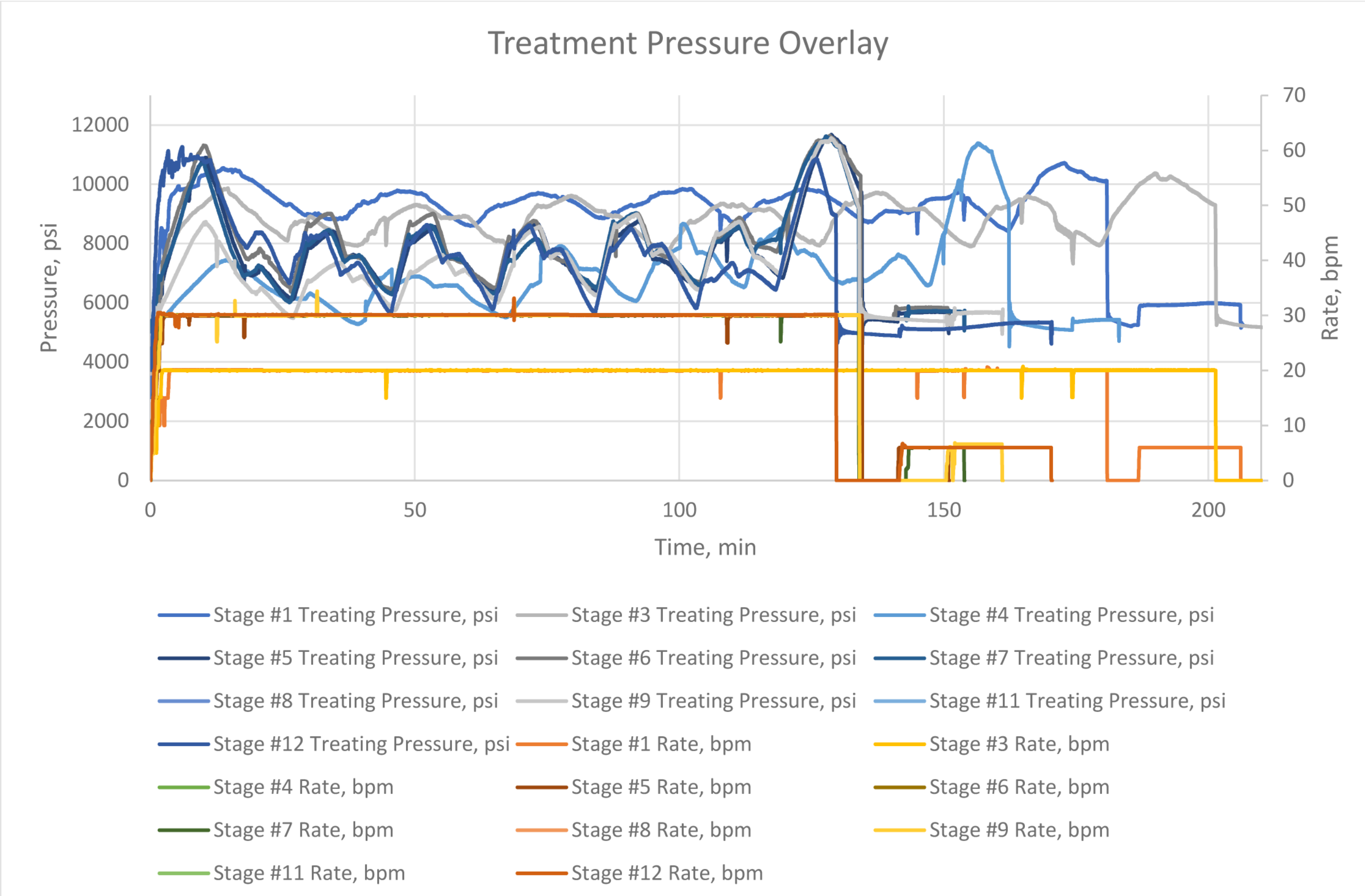

Figure 20—Treatment Timeline with Pressure Overlay for All Stages

Post-Stimulation Evaluation and Lessons Learned

The acid fracturing treatment resulted in significant stimulation of the carbonate formation, as evidenced by both the pressure-data diagnostics during the job and preliminary post-job well behavior. Key formation response indicators include fracture geometry evolution, acid penetration depth, leakoff characteristics, and the effectiveness of fracture etching:

- *Fracture Propagation and Containment*: The relatively moderate net pressure observed (on the order of 200–300 psi during acid injection) suggests that the created fractures were not pressure-limited by unusually high stresses or barriers. In the DataFrac, a net pressure of ~262 psi was measured, and during the main frac stages the net treating pressure (treating pressure minus closure) stayed in a similar range or lower. This implies that the fractures propagated without encountering extreme stress contrasts – a favorable condition indicating the treatment achieved vertical containment within the target zone. If net pressure had risen dramatically stage by stage, it could indicate height growth into higher stress layers; instead, pressures slightly dropped in later stages, implying fractures remained in-zone and possibly widened due to acid erosion.
- *Acid Etching and Conductivity*: The formation clearly responded to the acid by developing etched channels in the fracture faces. Evidence for this comes from the pressure falloffs during acid injection. Each time acid entered a fracture, the treating pressure fell. This reflects a rapid increase in fracture conductivity: as acid created wormholes and enlarged pore throats in the carbonate, the fracture faces offered less resistance to flow, so pressure decreased for the same injection rate. This

phenomenon was consistently observed, indicating successful acid-rock reaction. Additionally, the single-phase retarded acid has prolonged the reaction time, allowing acid to travel further down the fracture. Achieving longer etched half-length compared to using straight 20% HCl.

- *Leakoff and Fluid Efficiency*: The DataFrac analysis indicated a certain leakoff profile for the pad fluid, which was incorporated into the acid design. The addition of viscoelastic diverting acid (VDA) and the presence of gel filter cake from the pad helped control leakoff during the main treatment. The fluid efficiency (the fraction of fluid that remains in the fracture versus leaked off) was relatively high for an acid frac. We can infer this because the pressure did not precipitously fall during pumping – a sharp pressure fall would indicate excessive leakoff. Instead, pressures were sustained, meaning the fracture kept opening and held fluid. The VDA contributed to leakoff control by temporarily plugging pores as it viscosified. the treatment analysis indicated an apparent leakoff coefficient on the order of 0.001–0.002 ft/min^0.5 (derived from fall-off data), which is moderate for a carbonate and indicates some natural fractures or vugs taking fluid but not excessively. This manageable leakoff allowed the majority of the acid to be used for etching rather than being wasted into the formation matrix. Furthermore, the use of iron control agents and chelants ensured that any ferrous ions from spent acid did not precipitate as gels in the formation, which can plug pore throats.
- *Post-Treatment Indications*: After the job, the well was shut in for the acid to spend completely and the fractures to partially close. A step-down test and a brief flowback were performed. The step-down (pressure decay during pump shut-down) indicated minimal tortuosity or near-wellbore friction – a smooth pressure drop was recorded, meaning the perforations and fractures are likely free of damage or restrictions. On initial flowback, the well returned acid water and spent acid at a good rate, and within a short period began producing formation fluids. The wellhead pressure during flowback was higher than prior to stimulation, qualitatively confirming improved reservoir connectivity. Although a full production log is beyond the scope of this report, the early production data showed a significant increase (several-fold) in oil rate compared to pre-frac, in line with the treatment design expectations. This positive response can be attributed to the etched fractures providing high-conductivity paths from the reservoir to the wellbore. Any fines or damaged zones near the well were likely removed by the acid (with the mutual solvent assisting to clean emulsions), leaving a clean and highly permeable fracture face.

In conclusion, the formation responded very favorably to the acid fracturing treatment. The combination of customizable single phase deep-penetrating retarded acid and self-diverting acid stages resulted in long, well-distributed etched fractures. The pressure analysis during the job, corroborated by post-job well performance, indicates that the treatment objectives of enhanced fracture length and conductivity were achieved. The carbonate reservoir, initially characterized by high breakdown pressures and limited connectivity, was transformed into one with multiple draining fractures and significantly reduced flow resistance.

Discussion of Findings

The treatment provides several valuable insights and lessons learned for acid fracturing in similar carbonate reservoirs:

- *Performance of Retarded Acid vs Emulsified Acid:* The fully customizable single phase retarded acid with strength up to 28% used in this case appeared to perform excellently, achieving deep acid penetration without phase separation issues. Emulsified acids (acid-in-diesel emulsions) are a common method to retard acid reaction in high-temperature wells, but they come with drawbacks like sludge formation and higher pumping friction. The single-phase retarded system used here had lower friction and no hydrocarbon phase, which contributed to the smooth pumping operation achieving deep etching with a simpler fluid system. The lack of emulsion in the flowback and absence of persistent emulsified sludge indicates that this retarded acid was clean reacting. One observation is that the higher retarder loading did succeed in extending the acid's live distance. Given the results, the single- phase system was adequate and simpler to handle on location (mixing was straightforward, which minimized operational risk).

- *Efficacy of Self-Diverting Acid (VES-based):* The use of VDA (viscoelastic diverting acid) was pivotal for the treatment. Traditional acidizing often suffers from uneven acid placement – acid will preferentially go into the highest permeability streak or largest fracture and bypass other areas. In this case, the self-diverting property of the fluid clearly improved stimulation uniformity. Whenever one fracture path became dominant, the acid in that path quickly spent and gelled, forcing new acid into another path. The field evidence was the recurring pressure moderation during each acid stage, as opposed to a continual climb that one would expect if a single fracture was taking all acid. This confirms literature observations that VES-based acids can enhance zonal coverage and fluid efficiency in carbonate stimulations. An additional benefit observed was better leakoff control – effectively acting like a temporary blocking agent in the pores (as the VES viscosifies). Using a VES acid system is recommended when multiple perforations or high-perm contrasts are present. It should be noted that fluid compatibility and temperature stability of the VES should be verified for each case (here, the high-temp stability was addressed by additives and was sufficient for the bottomhole temperature.
- *Pressure Monitoring and Diagnostics:* The continuous recording of high-resolution pressure data proved crucial in understanding the formation response. By analyzing the minute-by-minute pressure changes, events such as fracture breakdown, acid spending, and diversion could be identified in real time. For example, the detection of the slight pressure dip during acid injection was only possible with careful monitoring and was a clear sign of acid doing its job. This underscores the importance of having a skilled pressure monitoring team and possibly automated pressure analysis tools (with fracturing software flagging diversion events or a typical pressure trend immediately). In a technical sense, the net pressure values and trends can be fed back into fracture models to update our understanding of fracture geometry. In this Treatment, the relatively flat net pressure across stages suggested similar fracture widths were achieved in each zone – a good outcome, as it means each zone was equivalently stimulated. If one zone had shown a much higher net pressure, it might indicate a narrower fracture or screen-out risk, which was not the case here. The treatment demonstrates how integrating data-driven design (via DataFrac), and real-time adaptive execution can overcome challenges of heterogeneity in carbonate reservoirs.
- *Operational Considerations:* From an operational perspective, a few discussion points emerge. The blending of multiple additives (guar, VES, retarder, etc.) in real time requires diligent QA/QC. We were able to keep the fluid within spec (no quality concerns reported) – this was aided by having on-site lab capabilities to adjust polymer concentration and crosslink timing on the fly. Pumping a corrosive fluid like HCl at high rates and pressures also demands attention to safety and equipment protection. The use of corrosion inhibitors and an inhibitor aid was essential for the long pump time. surface lines were kept full of inhibited pad or flush fluid whenever pausing, to avoid exposing iron to strong acid. These practices should be standard in acid fracs to prevent tubular damage. No indications of tubular corrosion were found in post-job inspections, validating the chemical program. Another operational note is the temperature of fluids: pumping such a large volume of fluid can cool the formation. The mutual solvent in the flush helped ensure that any emulsions from mixed oil/acid would break, even at possibly lower temperatures. Given the reservoir's heat, cooling was transient, but this factor should be considered in extended pumping operations (long

pumping times can reduce acid reactivity as the rock cools – however, retarded acid partly mitigates this by reaction kinetics rather than diffusion control).

- *Comparison to Expectations:* The outcomes of this acid fracturing treatment can be benchmarked against either an alternative approach or pre-job expectations. Prior to the job, modeling predicted a certain post-frac productivity index. Early production suggests we are meeting or exceeding that target. The uniform coverage likely played a role in achieving near the maximum potential stimulation. Had no diverting acid been used, some zones might have remained under stimulated, and the production gain would be less.
- *Environmental and Safety Aspects:* Acid fracturing involves handling large volumes of hazardous acid. From a safety standpoint, the job was executed without incident – no acid spills, and all personnel followed HSE protocols (acid suits, neutralizing agents on standby, etc.). On the environmental side, the use of modern additives like low-toxicity surfactants and optimized acid volume (the retarded acid allowed us to reduce total acid volume by ~30% compared to straight acid designs) helped minimize waste. Flowback handling included neutralizing the spent acid water before disposal. These considerations are increasingly important in today's operational environment.

Conclusions

The acid fracturing treatment achieved its objectives of enhanced reservoir stimulation and offered several conclusions for acid fracture design and execution in carbonate formations:

1. **Multi-Stage Acid Fracturing was Successfully Implemented**: Using a combination of fully customizable single phase retarded acid, and self-diverting acid in the treatment stimulated all targeted zones in a single continuous operation. Distinct fractures were created in six different intervals, as evidenced by separate breakdown events and pressure responses for each stage. This demonstrates that a properly engineered diversion strategy can enable effective multi-zone acid fracturing without the need for packers between zones.
2. **Viscoelastic Diverting Acid Improved Zonal Coverage:** The self-diverting acid played a crucial role in distributing acid across the fracture face and between multiple perforation clusters within each zone. The in-situ viscosity rise upon acid spending diverted fluid to less reactive areas, resulting in more uniform etching. Field data showed pressure during VDA injection, indicating reduced resistance and increased fracture conductivity as new surfaces were reached. The use of VES-based acid is concluded to significantly enhance acid placement efficiency in heterogeneous carbonates.
3. **Retarded Acid System Achieved Deep Etching:** The single-phase retarded acid allowed live acid to penetrate further into the fracture before spending, yielding longer etched fractures than conventional acid. Despite the high bottomhole temperature (~>200 °F), the retarder effectively slowed the reaction, as indicated by sustained pressure during injection and improved post-frac performance. The retarded acid over performed the more hazardous emulsified systems, confirming that such fluid type can provide deep penetration with simpler logistics. No emulsions or sludge were observed, and the retarded acid was pumped without elevated friction, simplifying the operation.
4. **Pressure Data Indicates Improved Fracture Conductivity and Containment:** The moderate net treating pressures and their decline in later stages suggest that the acid effectively enlarged the fractures (lowering fracture resistance) and that fracture growth remained within the desired zone. There was no sign of pressure drop that would hint at height growth into undesired layers; each stage's pressure profile was well-behaved. The consistent closure gradient observed before and after stimulation implies the fractures stayed in-zone and created extensive etched surface area without encountering barriers or breaking out-of-zone.

5. **Operational Execution Met Design Requirements:** The pumping equipment operated at the necessary high pressure and rate with no major issues. The fluids were mixed on-the-fly to the specified concentrations and remained stable. Corrosion control measures protected the well tubulars – no indications of tubing corrosion were found after the job. The treatment volumes and schedule were executed as planned, with only minor adjustments, demonstrating that complex acid blends can be reliably pumped with modern fracturing equipment and on-site quality control.
6. **Significant Well Performance Improvement:** Although detailed production data will be covered in separate reports, initial results are very positive. The post-frac well flow rates and pressures point to a substantial reduction in skin and increased productivity, validating the effectiveness of the acid fracturing treatment. The etched fractures have provided high-conductivity paths, and the removal of near-wellbore damage has likely improved inflow. This success case reinforces acid fracturing as a stimulation technique in similar carbonate reservoirs where conventional proppant fracturing is challenging or less effective.

In conclusion, this acid fracturing treatment case study illustrates the successful synergy of data-driven design, advanced fluid chemistries, and careful execution to overcome the challenges of stimulating a multi-layer carbonate formation. The methodologies and lessons learned here can be applied to future wells to maximize stimulation coverage and well performance in carbonate reservoirs.

Recommendation

Based on the outcomes and observations from this acid fracturing treatment, the following recommendations are made for future operations of a similar nature:

- **Employ Diversion Strategies:** Use chemical self-diverting acids in carbonate fracturing for optimal results. VES-based diverting acid is recommended to ensure uniform etching within each zone and is advisable for other wells with multiple target layers.
- **Calibrate with DataFrac/Injection Test:** Continue to perform a DataFrac or step-rate injection test prior to the main treatment. The closure pressure, leakoff coefficient, and efficiency obtained are invaluable for designing acid volumes and pump rates. In the treatment, the DataFrac data guided the retarder loading and VDA volume; similar calibration should be standard practice, especially in new fields or where in-situ stress is uncertain. If the DataFrac reveals very high leakoff or unexpectedly low closure stress, design adjustments can be made proactively.
- **Fluid System Considerations:** Maintain the use of single-phase retarded acid systems for deep, hot carbonate stimulations to simplify operations and minimize emulsions compared to emulsified acids. However, ensure thorough lab testing of the retarded acid at reservoir conditions (temperature, pressure, and live-fluid exposure) for each application. Verify that the retarder provides the needed reaction delay and that corrosion inhibitor packages are effective for the anticipated pump time at bottomhole temperature. Also, fine-tune the viscosity of pad fluids – enough to create fracture width but not too high to cause excessive friction or filter-cake that could impede acid entry.
- **Real-Time Monitoring and Adaptive Execution:** It is recommended to utilize real-time data analysis (perhaps via software or on-site experts) during acid fracturing. Subtle pressure trends should be interpreted on the fly to make decisions. Having a protocol for actions on certain pressure signatures is useful: Using downhole pressure gauges or fiber optics if available, to get more insight into fracture propagation and diversion effectiveness along the wellbore. These can validate whether all zones are fractured simultaneously or sequentially as intended.
- **Post-Frac Flowback and Cleanup:** After acid fracturing, a controlled flowback strategy is advised to maximize cleanup. Given the large acid volumes, gradually flowing the well helps remove spent

acid and fines without risk of collapsing fractures. Mutual solvent and proper breaking of gels results in a clean flowback. Monitoring of flowback pH and iron content is recommended; if spent acid returns have high iron or low pH after a certain duration, it may indicate incomplete spending or corrosion – in which case additional neutralization or inhibition steps might be taken.

- **Further Evaluation:** It is recommended to perform a post-stimulation well test or production log to quantify the contribution of each zone and verify that all zones are indeed producing. While the pressure responses indicated each was fractured, production logging can confirm zonal inflow split. This can validate the diversion success and guide any future stimulations (for example, if one zone under-produces, it can be targeted with a remedial treatment or future adjustment of perforation strategy. Additionally, gathering data on the longevity of stimulation (e.g., does the well exhibit sustained productivity increase over months) will inform the long-term effectiveness of the acid etching. Carbonate acid fractures can gradually lose conductivity if scale or fines redeposit – hence, consider scale inhibitor squeeze or acid additives that minimize $CaCO_3$ re-precipitation.
- **Knowledge Transfer:** Finally, it's recommended to document best practices for the operating company's stimulation programs. The design and execution learnings (e.g., optimal additive concentrations, diversion timing, pressure interpretation techniques) should be shared with other asset teams working in similar reservoirs. A workshop or technical memo can disseminate how treatment was engineered and how challenges were overcome, thereby improving the outcomes of subsequent acid fracturing jobs. By implementing these recommendations, acid fracturing operations can achieve effective, and safe stimulation of carbonate reservoirs with complex multi-zone targets.

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